

Exp no :01

Date :

Evidence Acquisition Using Access Data FTK Imager

Description

Forensic Toolkit or FTK is a computer forensics software product made by AccessData. This is a Windows based commercial product. For forensic investigations, the same development team has created a free version of the commercial product with fewer functionalities. This FTK Imager tool is capable of both acquiring and analyzing computer forensic evidence.

The evidence FTK Imager can acquire can be split into two main parts. They are:

- Acquiring volatile memory
- Acquiring non-volatile memory (Hard disk)

There are two possible ways this tool can be used in forensics image acquisitions:

- Using FTK Imager portable version in a USB pen drive or HDD and opening it directly from the evidence machine. This option is most frequently used in live data acquisition where the evidence PC/laptop is switched on.
- Installing FTK Imager on the investigator's laptop.

In this case the source disk should be mounted into the investigator's laptop via write blocker. The write blocker prevents data being modified in the evidence source disk while providing read-only access to the investigator's laptop. This helps to maintain the integrity of the source disk.

Acquiring volatile memory using FTK Imager

The FTK Imager tool helps investigators to collect the complete volatile memory (RAM) of a computer. The following steps will show you how to do this.

Open FTK Imager and navigate to the volatile memory icon (capture memory).

Memory Capture

Destination path:

C:\Drivers\FTK_MAGER

Browse

Destination filename:

memdump.mem

☒ Include pagefile

pagefile.sys

☒ Create AD1 file

memcapture.ad1

Capture Memory

Cancel

Memory Progress

Destination:

C:\Drivers\FTK_MAGER\memdump.mem

Status:

Dumping RAM: 16GB/19GB [83%]

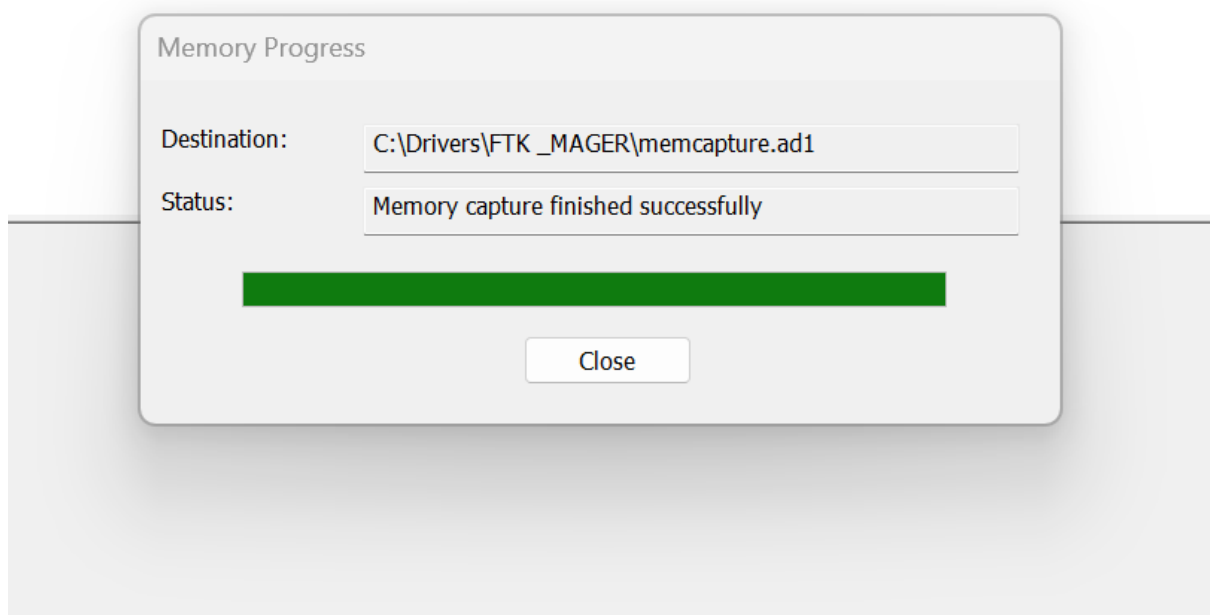
Cancel

NOTE: This tool provides options to include pagefile and AD1 files when acquiring the volatile memory.

Pagefile: The pagefile (pagefile.sys) is used in Windows operating systems as volatile memory due to limitation of physical random-access memory (RAM). It is located under the “C” partition ready to use as volatile memory when the existing RAM capacity is exceeded. So this file can have quite a bit of valuable data when considering the volatile memory. Therefore, it is recommended to capture and collect this file in the acquisition.

AD1 file: AD1 is the FTK imager image file. The investigator has the option to create an AD1 file for later use.

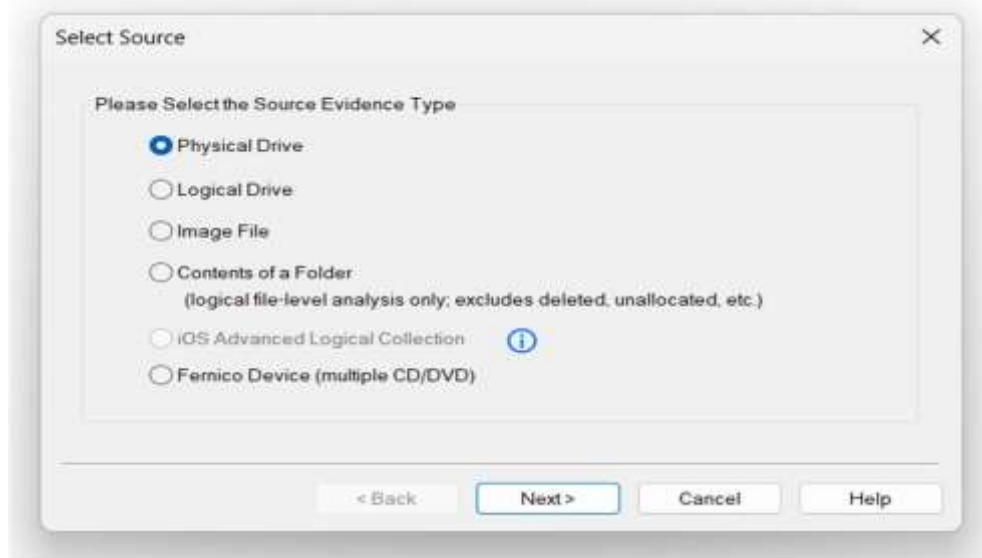
Clicking the “capture memory” button will start acquiring the volatile memory.



NOTE: Once the acquisition has completed, the destination folder will have the acquired memory with the file extension of “.mem”.

Acquiring non-volatile memory (Disk Image) using FTK Imager

As previously stated, this same tool can be used to collect a disk image as well. Open FTK Imager and navigate to “Create Disk Image”.



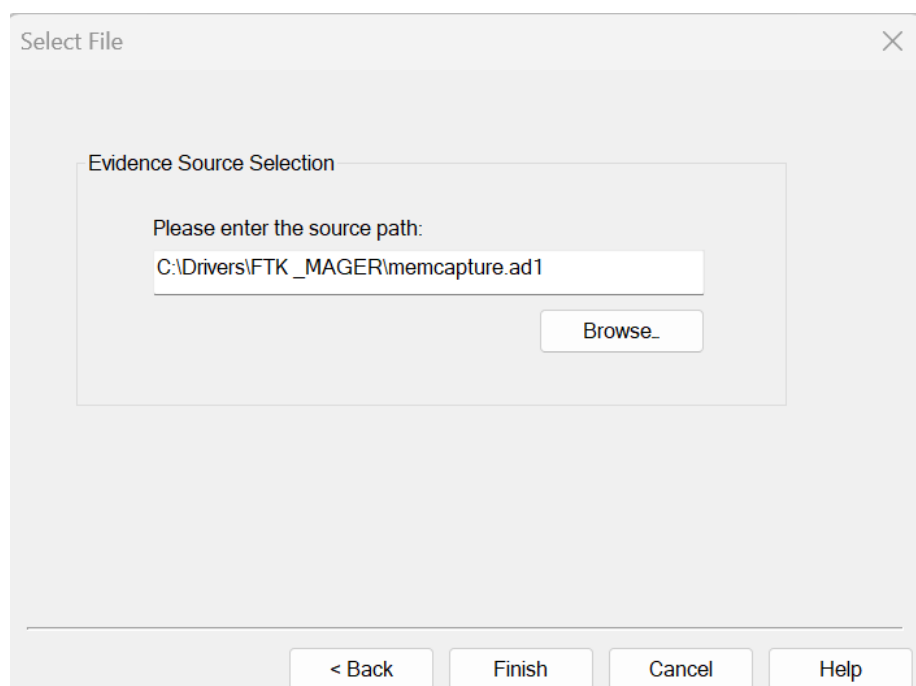
NOTE: FTK Imager is capable of acquiring physical drives (physical hard drives), logical drives (partitions), image files, contents of a folder, or CDs/DVDs. Investigators can connect external HDDs into the collection computer

via write blocker and use the “logical drive” option to select the mounted HDD as a partition.

Collecting Physical Drives

Select the “Physical Drive” option.

Select the drive you need to acquire and click “Finish”.



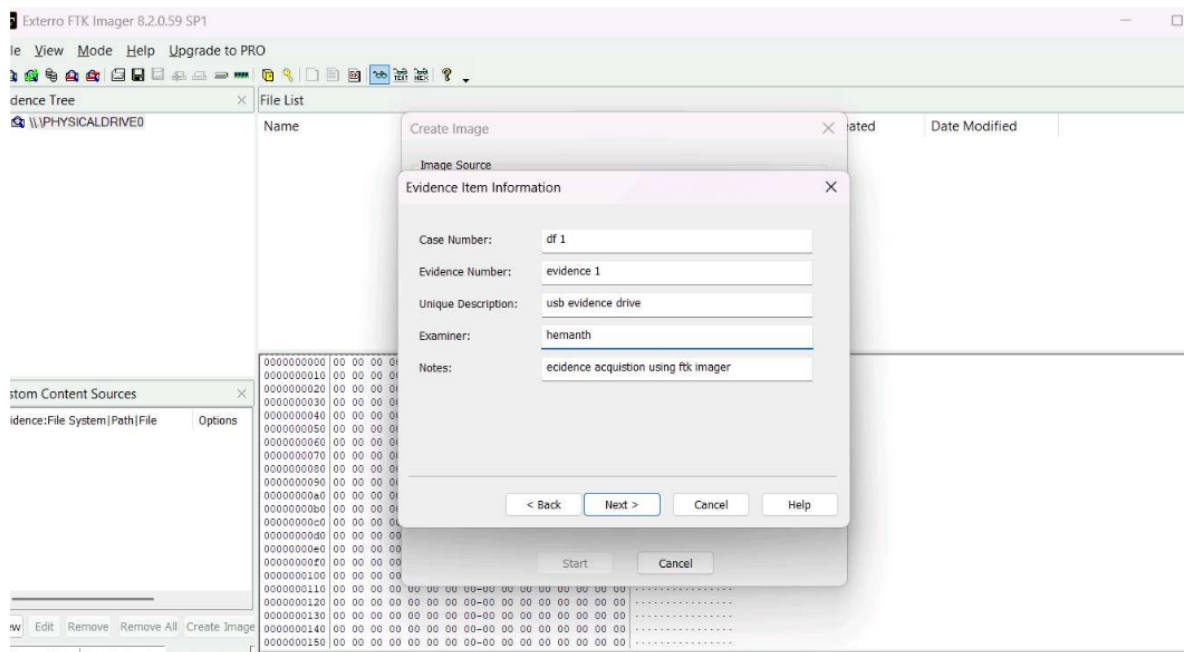
Raw (dd): This is the image format most commonly used by modern analysis tools. These raw file formatted images do not contain headers, metadata, or magic values. The raw format typically includes padding for any memory ranges that were intentionally skipped (i.e., device memory) or that could not be read by the acquisition tool, which helps maintain spatial integrity (relative offsets among data).

SMART: This file format is designed for Linux file systems. This format keeps the disk images as pure bitstreams with optional compression. The file consists of a standard 13-byte header followed by a series of sections. Each section includes its type string, a 64-bit offset to the next section, its 64-bit size, padding, and a CRC, in addition to actual data or comments, if applicable.

E01: this format is a proprietary format developed by Guidance Software's EnCase. This format compresses the image file. An image with this format starts with case information in the header and footer, which contains an MD5 hash of the entire bit stream. This case information contains the date and time of acquisition, examiner's name, special notes and an optional password.

AFF: Advance Forensic Format (AFF) was developed by Simson Garfinkel and Basis Technology. Its latest implementation is AFF4. The goal is to create a disk image format that does not lock the user into a proprietary format that may prevent them from being able to properly analyze it.

Now enter the case details.



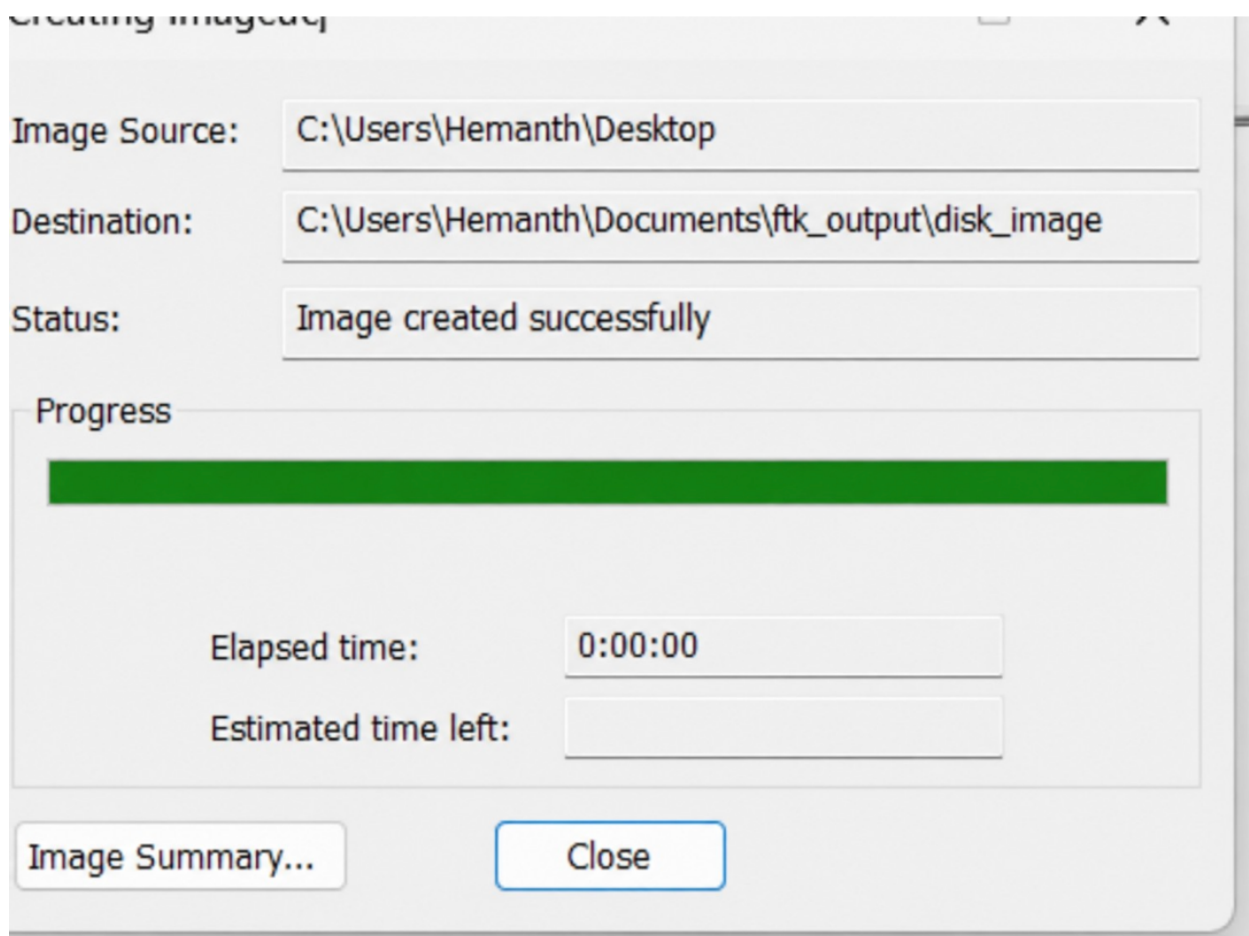
Add an image destination (where the image file will be saved), image file name and fragment size.

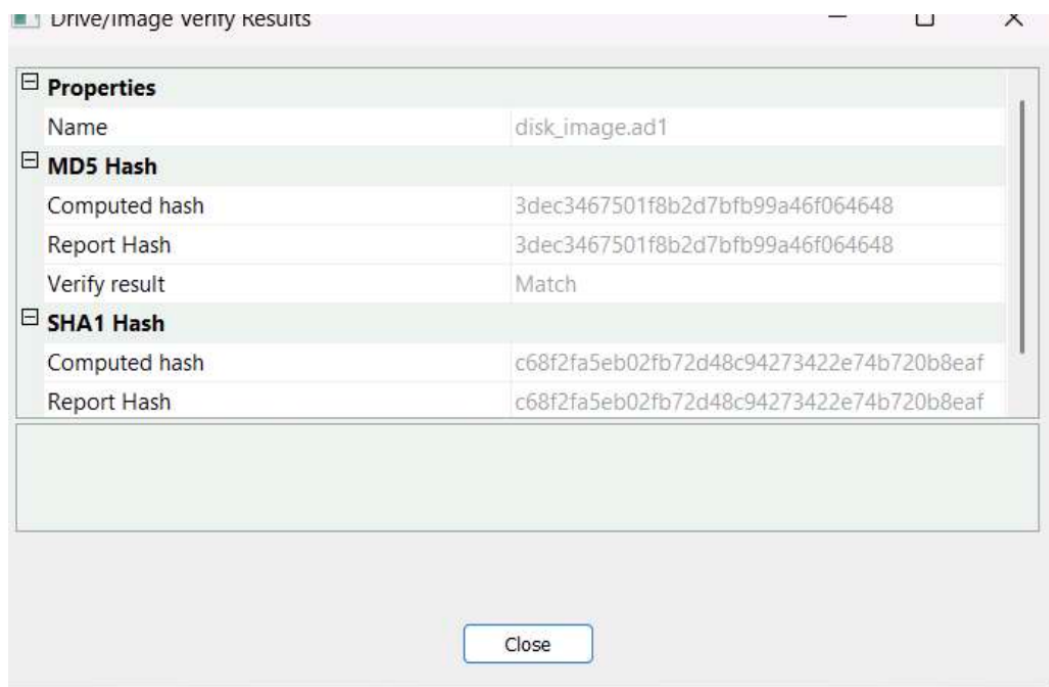
Image Fragment Size (MB): this option will separate the image file into multiple images and save them in the same destination. If you need only one file instead of creating multiple fragmented images, you must set the image fragment size to “0”.

Select the “verify images after they are created” option. This will verify the hash values once the image has created. In order to ensure integrity, it is recommended to use this option. However, this will increase the time taken to acquire your evidence, especially if you’re dealing with a large disk image size.

Click “start” to start acquiring.

Once acquiring is complete, it will create a text file including all the information it has acquired.





Hash values are matched.

Exp no :02

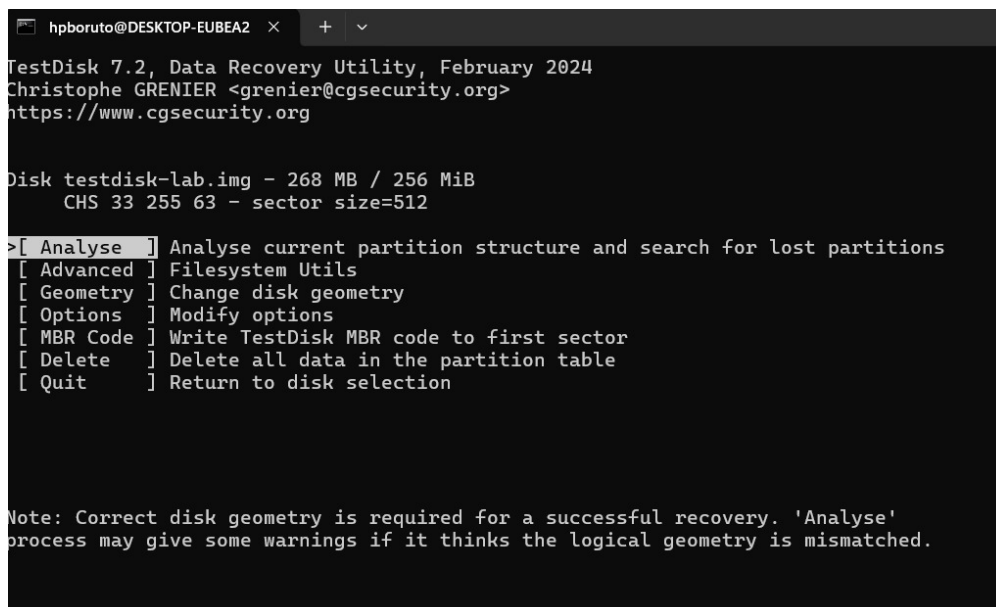
Date :

Recover deleted or damaged files from a storage device using Test Disk

Description:

TestDisk step by step to recover a missing partition and repair a corrupted one.

Log creation



```
hpboruto@DESKTOP-EUBE2
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk testdisk-lab.img - 268 MB / 256 MiB
CHS 33 255 63 - sector size=512

> [ Analyze ] Analyze current partition structure and search for lost partitions
  [ Advanced ] Filesystem Utils
  [ Geometry ] Change disk geometry
  [ Options ] Modify options
  [ MBR Code ] Write TestDisk MBR code to first sector
  [ Delete ] Delete all data in the partition table
  [ Quit ] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyze'
process may give some warnings if it thinks the logical geometry is mismatched.
```

- Choose Create to instruct Testdisk to create a [log file](#) containing technical information and messages, unless you have a reason to append data to the [log](#) or you execute TestDisk from read only media and must create the [log](#) elsewhere.
- Choose None if you do not want messages and details of the process to be written into a [log file](#) (useful if for example Testdisk was started from a read-only location).
- Press Enter to proceed.

Disk Selection

All hard drives should be detected and listed with the correct size by TestDisk:

```
hpboruto@DESKTOP-EUBE2  ×  +  v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk testdisk-lab.img - 268 MB / 256 MiB - CHS 33 255 63
Current partition structure:
    Partition              Start      End    Size in sectors

No partition is bootable

*=Primary bootable  P=Primary  L=Logical  E=Extended  D=Deleted
>[Quick Search]
                                Try to locate partition
```

- Use up/down arrow keys to select your hard drive with the lost partition/s.
- Press Enter to Proceed.

If available, use raw device /dev/sda* for faster data transfer.

Partition table type selection

TestDisk displays the partition table types.

```
hpboruto@DESKTOP-EUBEA2 x + v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk testdisk-lab.img - 268 MB / 256 MiB - CHS 33 255 63
Partition      Start      End      Size in sectors
>D Linux        0 32 33    8 40 32    129024
D Linux         0 32 33   32 162 2    522240

Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
*=Primary bootable P=Primary L=Logical E=Extended D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
ext4 blocksize=4096 Large_file Sparse_SB, 66 MB / 63 MiB
```

- Select the partition table type - usually the default value is the correct one as TestDisk auto-detects the partition table type.
- Press Enter to Proceed.

Current Partition Table Status

TestDisk displays the menus (also see [TestDisk Menu Items](#)).

```
hpboruto@DESKTOP-EUBE2  X  +  v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk testdisk-lab.img - 268 MB / 256 MiB
CHS 33 255 63 - sector size=512

>[ Analyse ] Analyse current partition structure and search for lost partitions
[ Advanced ] Filesystem Utils
[ Geometry ] Change disk geometry
[ Options ] Modify options
[ MBR Code ] Write TestDisk MBR code to first sector
[ Delete ] Delete all data in the partition table
[ Quit ] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyse'
process may give some warnings if it thinks the logical geometry is mismatched.
```

- Use the default menu "Analyse" to check your current partition structure and search for lost partitions.
- Confirm at Analyse with Enter to proceed.

Now, your current partition structure is listed. Examine your current partition structure for missing partitions and errors.

```
hpboruto@DESKTOP-EUBEA2 × + v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk testdisk-lab.img - 268 MB / 256 MiB - CHS 33 255 63

Partition                                Start      End      Size in sectors
1 E extended                            0 32 32    8 254 63    142538
5 L Linux                               0 32 33    8 40 32    129024

[ Quit ] [ Return ] >[Deeper Search] [ Write ] [Extd Part]
                        Try to find more partitions
```

The first partition is listed twice which points to a corrupted partition or an invalid partition table entry.

Invalid NTFS boot points to a faulty NTFS boot sector, so it's a corrupted filesystem.

Only one logical partition (label Partition 2) is available in the extended partition. One logical partition is missing.

- Confirm at **Quick Search** to proceed.

Quick Search for Partitions

During the **Quick Search**, TestDisk has found two partitions including the missing logical partition labeled *Partition 3*.

- Highlight this partition and press **p** to list your files (to go back to the previous display, press q to Quit, Files listed in red are deleted entries).

All directories and data are correctly listed.

- Press Enter to proceed.

Save the Partition table or search for more partitions

```
hpboruto@DESKTOP-EUBE2  ×  +  v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk testdisk-lab.img - 268 MB / 256 MiB - CHS 33 255 63
Partition      Start      End      Size in sectors
>L Linux        0  32 33    8  40 32    129024
D Linux        0  32 33   32 162  2    522240

Structure: Ok.  Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
*=Primary bootable  P=Primary  L=Logical  E=Extended  D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
ext4 blocksize=4096 Large_file Sparse_SB, 66 MB / 63 MiB
```

- *When all partitions are available* and data correctly listed, you should go to the menu **Write** to save the partition structure. The menu Extd Part gives you the opportunity to decide if the extended partition will use all available disk space or only the required (minimal) space.
- *Since a partition, the first one, is still missing*, highlight the menu **Deeper Search** (if not done automatically already) and press Enter to proceed.

TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
<https://www.cgsecurity.org>

Disk testdisk-lab.img - 268 MB / 256 MiB - CHS 33 255 63

Partition	Start	End	Size in sectors
>D Linux	0 32 33	8 40 32	129024
D Linux	0 32 33	32 162 2	522240

Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
*=Primary bootable P=Primary L=Logical E=Extended D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
ext4 blocksize=4096 Large_file Sparse_SB, 66 MB / 63 MiB

```
hpboruto@DESKTOP-EUBEA2 x + v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
Linux 0 32 33 8 40 32 129024
Directory /Forensics
>drwxr-xr-x 0 0 4096 24-Aug-2026 06:21 .
drwxr-xr-x 0 0 4096 24-Aug-2026 06:20 ..
-rw-r--r-- 0 0 30 24-Aug-2026 06:20 file1.txt
-rw-r--r-- 0 0 29 24-Aug-2026 06:21 file2.txt

Next
Use Left arrow to go back, Right to change directory, 'h' to hide deleted files
'q' to quit, ':' to select the current file, 'a' to select all files
'C' to copy the selected files, 'c' to copy the current file
```

It works, your files are listed, you have found the correct partition!

- Use the left/right arrow to navigate into your folders and watch your files for more verification

Note: FAT directory listing is limited to 10 clusters - some files may not appear but it doesn't affect recovery.

- Press q for Quit to go back to the previous display.
- The available status are Primary, * bootable, Logical and Deleted.

Using the left/right arrow keys, change the status of the selected partition from Deleted to **Logical**. This way you will be able to recover this partition.

```
hpboruto@DESKTOP-EUBEA2  ×  +  v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk testdisk-lab.img - 268 MB / 256 MiB - CHS 33 255 63
Current partition structure:
    Partition                Start          End      Size in sectors
  1 E extended                0  32  32      8 254 63      142538
No partition is bootable
  5 L Linux                   0  32  33      8  40 32      129024

*=Primary bootable  P=Primary  L=Logical  E=Extended  D=Deleted
>[Quick Search]  [ Backup ]
                                Try to locate partition
```

TestDisk displays *You have to restart your Computer to access your data* so press Enter a last time and reboot your computer.

Exp no :03

Date :

WIRESHARK FOR PASSWORD CAPTURING

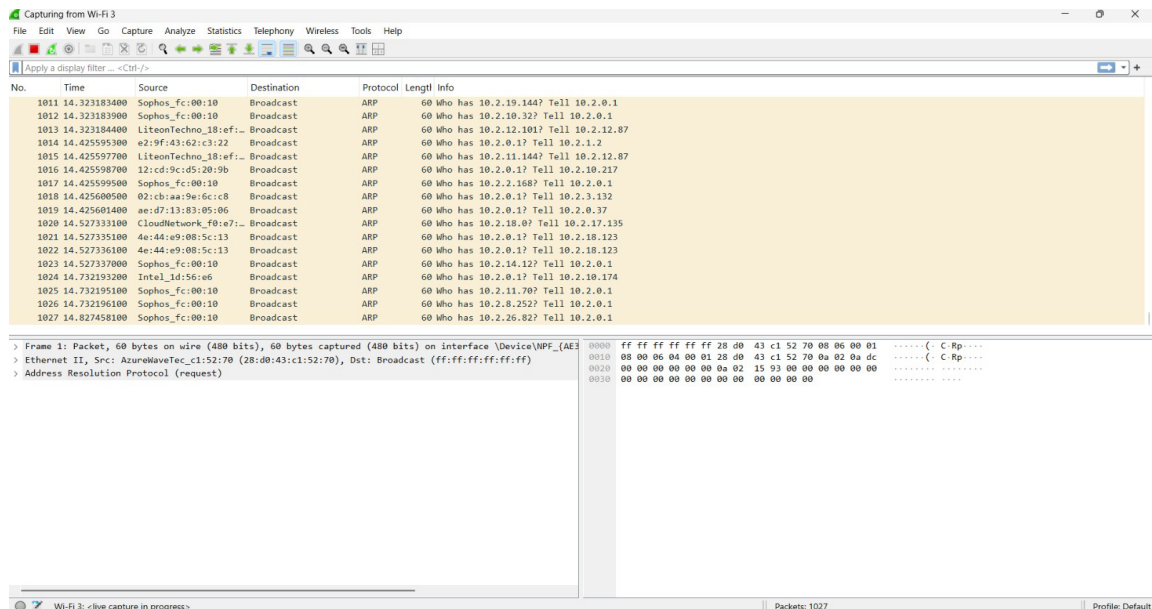
Password Capturing

Wireshark can capture not only passwords but any type of information transmitted over the network: usernames, email addresses, personal information, etc. As long as we can capture network traffic, Wireshark can sniff passing passwords.

In sniffing can include passwords for various protocols such as HTTP, FTP, Telnet, etc. the captured data can be used to troubleshoot network problems, but can also be used maliciously to gain unauthorized access to sensitive information.

So, here we will see how we can capture the password using the Wireshark network capture analyzer. and see the outputs of the following steps.

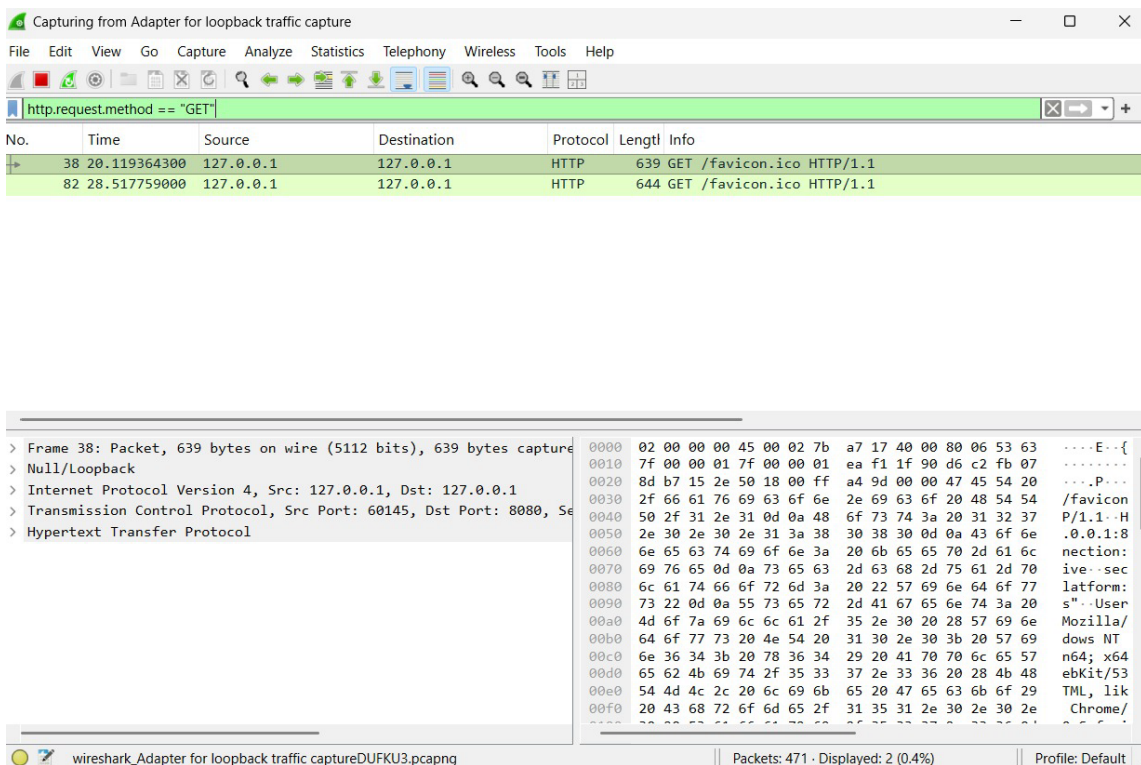
Step 1: First of all, open your Wireshark tool in your window or in Linux virtual machine. and start capturing the network. suppose I am capturing my wireless fidelity.



After starting the packet capturing we will go to the website and login the credential on that website as you can see in the image

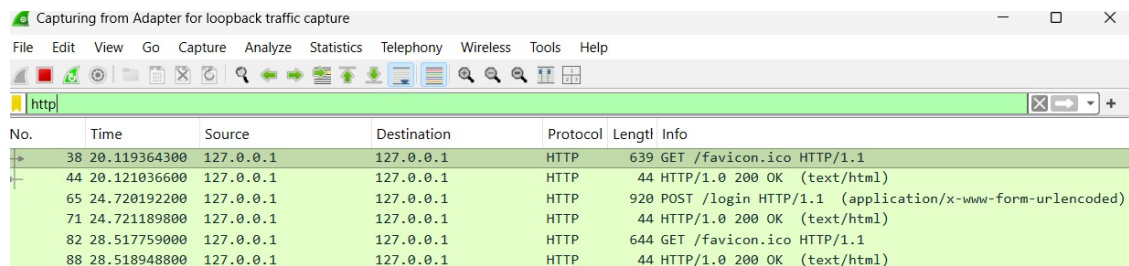
Step 3: Now after completing the login credential we will go and capture the password in Wireshark. for that we have to use some filter that helps to find the login credential through the packet capturing.

Step 4: Wireshark has captured some packets but we specifically looking for HTTP packets. so in the display filter bar we use some command to find all the captured HTTP packets. as you can http

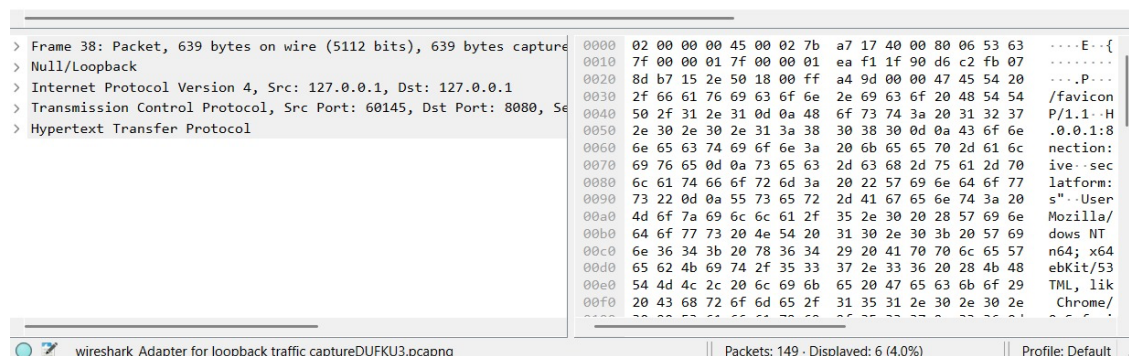


Step 5: So there are some HTTP packets are captured but we specifically looking for form data that the user submitted to the website. for that, we have a separate filter

As we know that there are main two methods used for submitting form data from web pages like login forms to the server. the methods are-GET



No.	Time	Source	Destination	Protocol	Length	Info
38	20.119364300	127.0.0.1	127.0.0.1	HTTP	639	GET /favicon.ico HTTP/1.1
44	20.121036600	127.0.0.1	127.0.0.1	HTTP	44	HTTP/1.0 200 OK (text/html)
65	24.720192200	127.0.0.1	127.0.0.1	HTTP	920	POST /login HTTP/1.1 (application/x-www-form-urlencoded)
71	24.721189800	127.0.0.1	127.0.0.1	HTTP	44	HTTP/1.0 200 OK (text/html)
82	28.517759000	127.0.0.1	127.0.0.1	HTTP	644	GET /favicon.ico HTTP/1.1
88	28.518948800	127.0.0.1	127.0.0.1	HTTP	44	HTTP/1.0 200 OK (text/html)



Frame 38: Packet, 639 bytes on wire (5112 bits), 639 bytes captured on interface Null/Loopback	
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1	
Transmission Control Protocol, Src Port: 60145, Dst Port: 8080, Seq: 300000000, Win: 65535, Len: 639	
Hypertext Transfer Protocol	
0000	02 00 00 00 45 00 02 7b a7 17 40 00 80 06 53 63E--{
0010	7f 00 00 01 7f 00 00 01 ea f1 1f 90 d6 c2 fb 07{
0020	8d b7 15 2e 50 18 00 ff a4 9d 00 00 47 45 54 20P...
0030	2f 66 61 76 69 63 6f 6e 2e 69 63 6f 20 48 54 54 /favicon
0040	50 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 31 32 37 P/1.1..H
0050	2e 30 2e 30 2e 31 3a 38 30 38 30 0d 0a 43 6f 6e .0.0.1:8
0060	6e 65 63 74 69 6f 6e 3a 20 6b 65 65 70 2d 61 6c nection:
0070	69 76 65 0d 0a 73 65 63 2d 63 68 2d 75 61 2d 70 ive..sec
0080	6c 61 74 66 6f 72 6d 3a 20 22 57 69 6e 64 6f 77 latform:
0090	73 22 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 s"..User
00a0	4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 28 57 69 6e Mozilla/
00b0	64 6f 77 73 20 4e 54 20 31 30 2e 30 3b 20 57 69 dows NT
00c0	6e 36 34 3b 20 78 36 34 29 20 41 70 70 6c 65 57 n64; x64
00d0	65 62 4b 69 74 2f 35 33 37 2e 33 36 20 28 4b 48 ebKit, lik
00e0	54 4d 4c 2c 20 6c 69 6b 65 20 47 65 63 6b 6f 29 TML, lik
00f0	20 43 68 72 6f 6d 65 2f 31 35 31 2e 30 2e 30 2e Chrome/

POST

Step 6: So firstly for knowing the credential we use the first method and apply the filter for the GET methods as you can see below.

`http.request.method == "GET"`

see in the below image the green bar where we apply the filter.

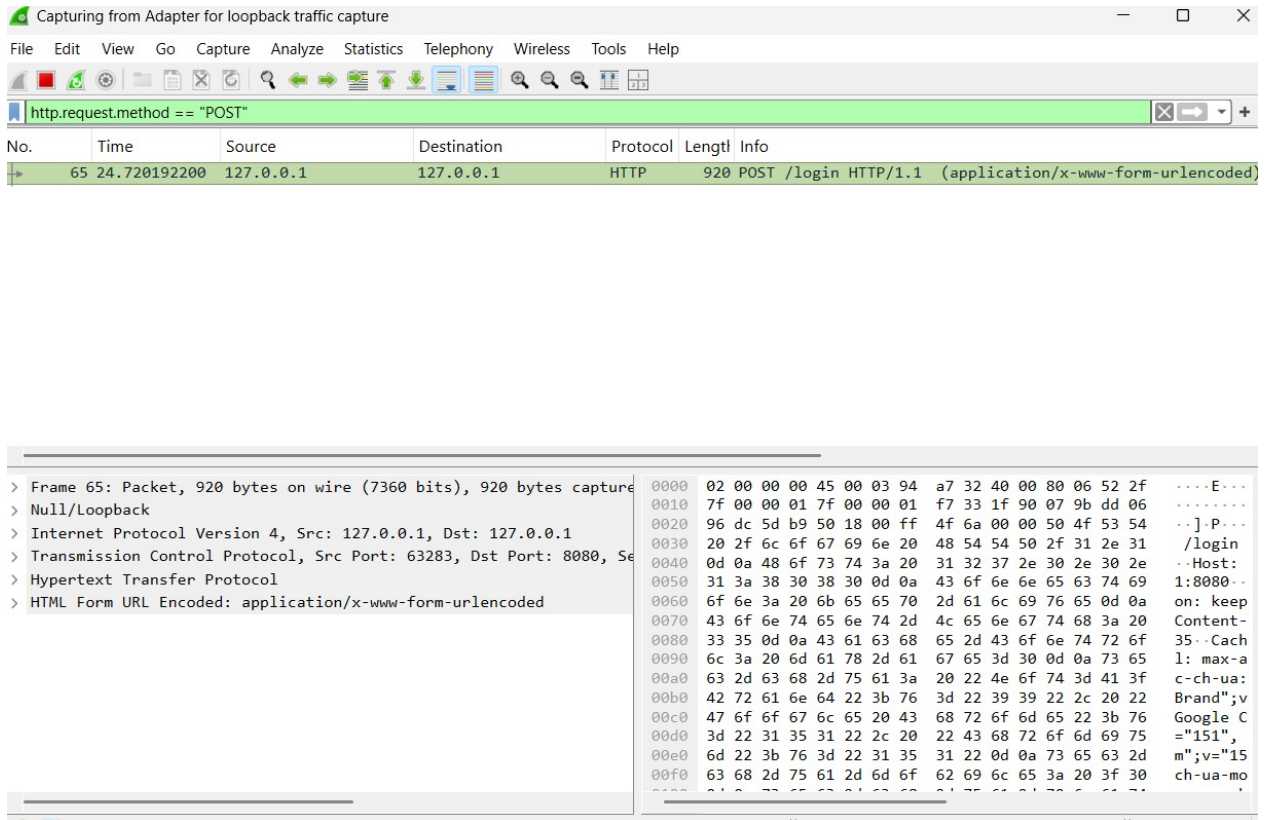
GET Method

As you can see in the image there are two packets where the login page was requested with a GET request as well, but

there is no form data submitted with a GET request.

Step 7: Now after checking the GET method if we didn't find the form data, then we will try the POST method for that we will apply the filter on Wireshark as you can see.

`http.request.method == "POST"`



As you can see we have a packet with form data click on the packet with user info and the application URL encoded. and click on the down-

HTML form URL Encoded where the login credential is found. login credential as it is the same that we filed on the website in step 2.

Form item: "username"="Hemanth"

Form item: "pass" = "Hemanth123\$"

Wireshark · Packet 65 · Adapter for loopback traffic capture

- > Frame 65: Packet, 920 bytes on wire (7360 bits), 920 bytes captured (7360 bits) on interface \Device\NPF
- > Null/Loopback
- > Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
- > Transmission Control Protocol, Src Port: 63283, Dst Port: 8080, Seq: 1, Ack: 1, Len: 876
- > Hypertext Transfer Protocol
- > HTML Form URL Encoded: application/x-www-form-urlencoded

0000	02 00 00 00 45 00 03 94	a7 32 40 00 80 06 52 2f	E... .2@...R/
0010	7f 00 00 01 7f 00 00 01	f7 33 1f 90 07 9b dd 06	-3.....
0020	96 dc 5d b9 50 18 00 ff	4f 6a 00 00 50 4f 53 54	..].P...Oj..POST
0030	20 2f 6c 6f 67 69 6e 20	48 54 54 50 2f 31 2e 31	/login HTTP/1.1
0040	0d 0a 48 6f 73 74 3a 20	31 32 37 2e 30 2e 30 2e	..Host: 127.0.0.
0050	31 3a 38 30 38 30 0d 0a	43 6f 6e 6e 65 63 74 69	1:8080.. Connecti
0060	6f 6e 3a 20 6b 65 65 70	2d 61 6c 69 76 65 0d 0a	on: keep -alive..
0070	43 6f 6e 74 65 6e 74 2d	4c 65 6e 67 74 68 3a 20	Content- Length:
0080	33 35 0d 0a 43 61 63 68	65 2d 43 6f 6e 74 72 6f	35..Cach e-Contro
0090	6c 3a 20 6d 61 78 2d 61	67 65 3d 30 0d 0a 73 65	l: max-a ge=0..se
00a0	63 2d 63 68 2d 75 61 3a	20 22 4e 6f 74 3d 41 3f	c-ch-ua: "Not=A?
00b0	42 72 61 6e 64 22 3b 76	3d 22 39 39 22 2c 20 22	Brand";v ="99", "
00c0	47 6f 6f 67 6c 65 20 43	68 72 6f 6d 65 22 3b 76	Google C hrome";v

No.: 65 · Time: 24.720192200 · Source: 127.0.0.1 · Destination: 127.0.0.1 · Protocol: HTTP · Length: 920 · Info: POST /login HTTP/1.1 (application/x-www-form-urlencoded)

☒ Show packet bytes Layout: Vertical (Stacked)

Close

Help

Exp no :04

Date :

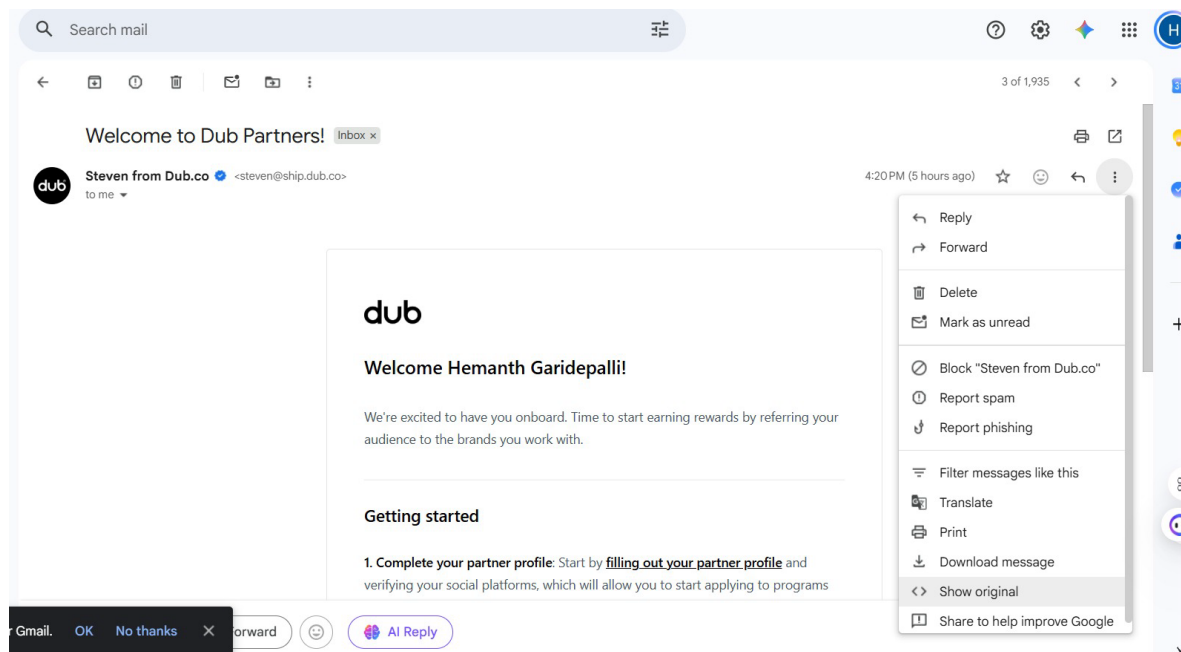
Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- Gmail: Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- Outlook: Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- Yahoo: Open the email, click on the three dots (More), and select "View raw message."



2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

Original Message

Message ID	<010001a07b7e1468-15797c73-0530-4671-bbc8-894a3a0f9131-000000@email.amazonses.com>
Created at:	Mon, Sep 7, 2026 at 4:20 PM (Delivered after 1 second)
From:	"Steven from Dub.co" <steven@ship.dub.co>
To:	garidepallihemanth@gmail.com
Subject:	Welcome to Dub Partners!
SPF:	PASS with IP 76.223.140.11 Learn more
DKIM:	'PASS' with domain ship.dub.co Learn more
DMARC:	'PASS' Learn more

[Download Original](#)

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - **The sending server's hostname/IP address.**
 - **The receiving server's hostname/IP address.**
 - **The date and time the email passed through.**

5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

Whois IP 76.223.140.11

Updated 1 second ago

```
#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#

# start

NetRange:      76.223.0.0 - 76.223.175.255
CIDR:          76.223.0.0/17, 76.223.128.0/19, 76.223.160.0/20
NetName:       AMAZO-4
NetHandle:     NET-76-223-0-0-1
Parent:        NET76 (NET-76-0-0-0-0)
NetType:       Direct Allocation
OriginAS:
Organization:  Amazon.com, Inc. (AMAZO-4)
RegDate:       2018-01-10
Updated:       2018-03-07
Ref:           https://rdap.arin.net/registry/ip/76.223.0.0
```

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.

7. Analyze the 'Message-ID'

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

Header Analyzed

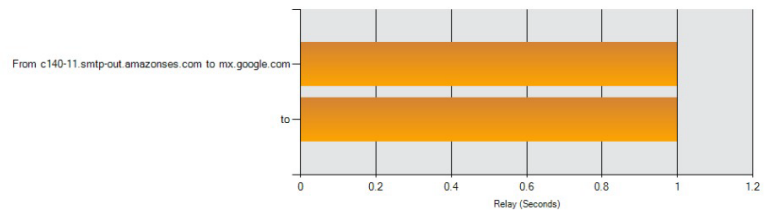
Email Subject: [Welcome to Dub Partners!](#)

Delivery Information

- ✓ DMARC Compliant
 - ✓ SPF Alignment
 - ✓ SPF Authenticated
 - ✓ DKIM Alignment
 - ✓ DKIM Authenticated

Relay Information

Received Delay: 0 seconds



SPF and DKIM Information

dmARC:ship.dub.co

Show

[Solve Email Delivery Problems](#)

v=DMARC1;p=quarantine;pct=100;rua=mailto:dmARCreports@dub.co

SPF:bounces.ship.dub.co:76.223.140.11

Show

[Solve Email Delivery Problems](#)

v=spf1 include:_spf.google.com include:amazonses.com ~all

DKIM:ship.dub.co:resend

Show

DKIM Public Record:

p=MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQDdckswZImaDU706y6oe9DXHBsTL2/FgDMgkBBj12kSbX8TIt5hBnoybhisi482ddly5ioHwFBVG081F6YMKW0unzhHezii+8544K1PDN33ww1B37VZR8bP0luBXz1JeZ0P7HDqG8M20RV3bwZTknXi

DKIM Signature:

v=1; a=rsa-sha256; q=dns/txt; c=relaxed/simple; s=resend; d=ship.dub.co; t=1788778256; h=List-Unsubscribe:From:To:Reply-To:Subject:Message-ID:Date:MIME-Version:Content-Type; bh=sy025m/Q7ln4w6X

Your IP is: 193.238.238.184 | [Contact](#) [Terms & Conditions](#) [Security](#) [API](#) [Privacy](#) Phone: (866) 408-4652 | © Copyright 2004-2025, MXToolbox, Inc. All rights reserved. US Patents 10039353 B2 & 11461738 B2

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.